

I build forecasts and optimisations that pay for themselves.

Demand plans, safety stock and resource allocation — engineered so the result shows up in cost, not in a leaderboard of accuracy metrics.

What I do

Demand & inventory

Forecasts at SKU level, and the replenishment rules that turn them into stock. Judged on what it costs to run, not on how close the line sits to the curve.

Constrained allocation

Who or what goes where, when budget, capacity and rules all bind at the same time. Solved to proven optimality rather than approximated and hoped over.

Decisions in production

Models that keep working when the data moves, with the failure modes designed in rather than discovered in production.

Selected work

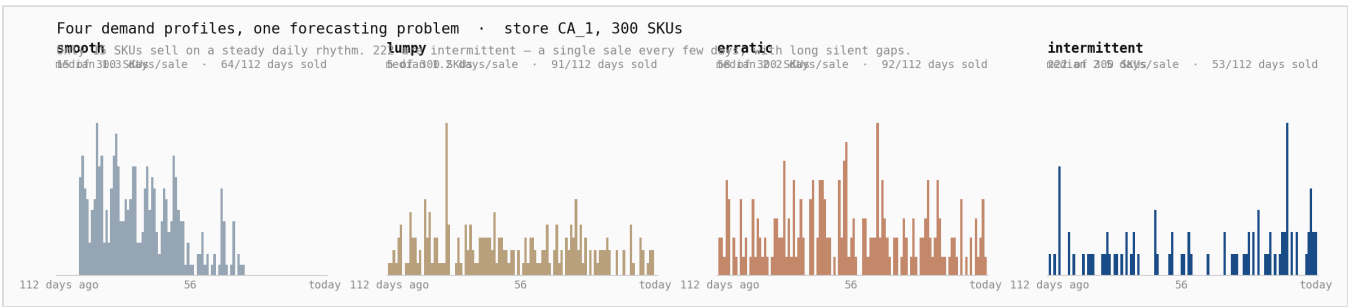
Three problems, what each one cost, and where the result stops being trustworthy.

Retail demand → inventory cost

Forecasting 300 SKUs that refuse to behave like each other

THE PROBLEM

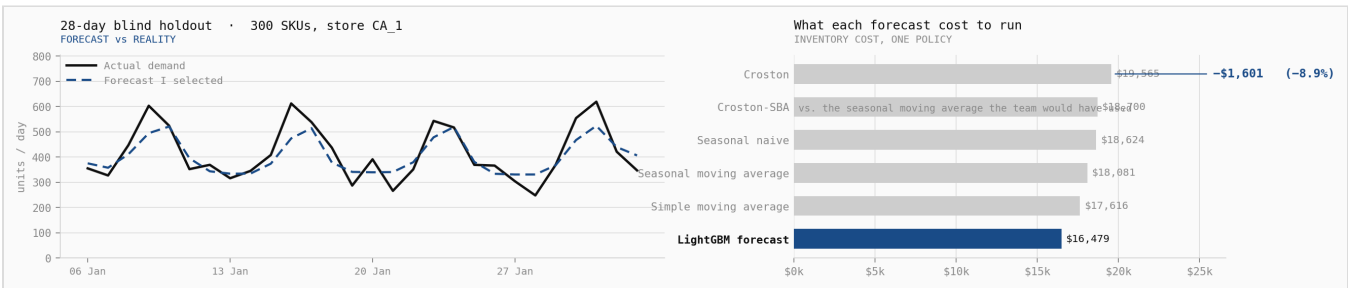
Retail demand isn't one curve. The same shelf holds SKUs that sell steadily every day and SKUs that sell two units a month in a single burst. A forecast tuned on the first kind quietly ruins the second — and in this catalogue, 74% of SKUs were the second kind.



Four demand profiles from the same store. Only 15 of 300 SKUs sell on a steady daily rhythm. 222 are intermittent — a single sale every few days, with long silent gaps that no smooth curve can express.

THE RESULT

On a blind 28-day holdout the model I selected cost **\$1,601 less to run** than the seasonal moving average a competent team would realistically have shipped — 8.9% of total inventory cost, while holding service level at 90.4%.



What the forecast actually cost to run. Every candidate is scored under one identical replenishment policy, so the comparison is the forecast's, not the policy's.

300 SKUs · 74% intermittent · **-\$1,601** inventory cost (-8.9%) · 90.4% fill rate · 11,585 units in window

THE CAVEAT

One store, one fixed replenishment policy, one 28-day window. Whether the same forecast still wins when the policy itself is tuned, or across more stores, is untested.

Constrained optimisation

Picking eleven people under rules that all bind at once

THE PROBLEM

Budget, formation balance, positional versatility and players already decided on — each is simple alone, and contradictory together. Scoring candidates by hand finds something plausible. It never finds the best one.

THE RESULT

Solved as a mixed-integer program, so the returned squad is provably the best available under the constraints, and a scenario that *cannot* be satisfied is reported as infeasible instead of quietly returning a bad team. Running now at squad-optimizer.streamlit.app.

THE CAVEAT

One objective, one decision, and results only as good as the player ratings feeding it. Real selection carries weights I can't infer from a spreadsheet.

Production ML

Keeping a classifier useful when the rare case is the one that matters

THE PROBLEM

With under 10% of records in the class that matters, a single decision threshold forces a bad trade: flood the review queue, or miss the very thing the model was built to catch.

THE RESULT

A three-tier confidence router — decide, escalate, or log for retraining — deployed as a service. It keeps returning usable output when the traffic drifts, and the escalation queue turns into its own training signal instead of a backlog someone has to clear.

THE CAVEAT

The text features won't transfer as they stand. The routing and calibration design is the part that travels to a different problem.

See it running

Not a slide deck. The forecasting work is a live dashboard over the holdout run.

It reads the same blind holdout the numbers above come from: the demand-pattern map across all 300 SKUs, inventory cost for every candidate forecast side by side, and a review-by-review trace of each replenishment decision the system made — including the ones it got wrong.

A written business report translating the same run into commercial terms is in preparation.

About

PhD in condensed matter physics at IPCMS, University of Strasbourg, spent on large finite-element simulations and the pipelines that made their output usable. The habit that carried over: build the baseline first, check it against something real, then report the one number the person deciding actually needs.

I work with clients on forecasting and optimisation problems where the data already exists and the impact is measurable. Based in Strasbourg, working remote or hybrid across the EU.

Peer-reviewed: `exact solution of the spin-1/2 Heisenberg model using Grassmann natural fermion operators` – Computer Physics Communications, 2023.

Ashfaque Thonikkadavan

CURRICULUM VITAE

Tell me the decision you're trying to improve.

If a simpler approach serves you better than what I would build, I'll say so on the first call.